

OCEANA NAS, VA, USA

WMO: 723075

Lat: 36.817N

Lon: 76.033W

Elev: 23

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13769

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.9	(c) 25.1	(d) 3.4	(e) 6.6	(f) 26.2	(g) 8.3	(h) 8.4	(i) 29.4	(j) 29.1	(k) 41.7	(l) 26.2	(m) 40.9	(n) 10.0	(o) 340	(p) 0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	93.0	77.1	90.6	76.3	88.4	75.4	79.4	89.5	78.1	87.2	77.2	85.4	10.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	138.9	83.9	75.4	133.2	82.5	74.7	129.9	81.8	42.8	89.4	41.5	87.3	40.5	85.5	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.8	(b) 21.2	(c) 19.0	(e) 15.1	(f) 97.4	(g) 5.0	(h) 2.4	(i) 11.5	(j) 99.2	(k) 8.6	(l) 100.6	(m) 5.9	(n) 101.9	(o) 2.2	(p) 103.7		
(a) 24.8	(b) 21.2	(c) 19.0	(e) 13.1	(f) 81.9	(g) 4.6	(h) 1.5	(i) 9.8	(j) 83.0	(k) 7.1	(l) 83.9	(m) 4.5	(n) 84.7	(o) 1.2	(p) 85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	41.4	43.5	49.7	59.0	67.0	75.3	79.3	78.0	72.6	62.8	52.1	45.6
	DBStd	15.51	10.47	9.50	9.53	8.74	7.49	6.01	4.63	4.48	5.20	7.71	8.60	9.55
	HDD50	958	308	227	127	16	0	0	0	0	0	3	73	204
	HDD65	3259	732	602	484	224	70	5	0	0	6	136	395	605
	CDD50	4839	42	46	117	286	528	760	910	867	678	401	137	67
	CDD65	1665	1	2	9	44	133	315	445	402	234	69	8	3
	CDH74	13969	1	11	80	361	990	2740	4369	3568	1440	364	30	15
	CDH80	4689	0	0	10	80	287	978	1703	1230	337	62	1	1
(14) Wind	WSAvg	9.2	10.0	10.2	10.6	10.7	9.5	8.6	7.9	7.4	8.5	8.6	8.9	9.1
(15) Precipitation	PrecAvg	44.2	3.8	3.4	3.7	2.8	3.5	3.5	5.0	5.0	3.9	3.4	2.6	3.0
	PrecMax	58.2	11.4	9.0	8.1	6.6	6.6	7.5	12.4	13.6	15.3	10.0	5.7	5.8
	PrecMin	28.1	1.2	0.7	0.9	1.0	0.4	0.8	0.9	1.4	0.2	0.2	0.1	0.6
	PrecStd	6.8	2.0	1.6	1.7	1.4	1.6	1.7	2.5	3.0	3.1	2.4	1.5	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.2	75.4	81.2	86.3	90.3	94.8	97.3	95.7	90.3	84.9	77.6	75.9
		MCWB	62.4	62.4	63.9	67.3	72.5	76.4	79.4	78.7	75.3	71.4	66.4	66.4
	2%	DB	67.5	69.6	75.4	82.2	86.5	91.2	93.4	91.8	86.9	81.5	73.1	70.1
		MCWB	60.5	58.9	61.6	66.3	71.6	75.2	77.8	77.4	74.3	70.6	64.4	63.2
	5%	DB	63.2	64.8	70.7	78.7	83.4	88.5	91.0	89.4	83.8	78.1	69.8	65.8
		MCWB	57.2	57.2	59.0	64.4	70.5	74.6	77.3	76.5	73.2	69.2	63.0	60.7
	10%	DB	58.8	60.1	65.8	73.7	80.0	85.7	88.4	86.4	81.2	74.9	66.1	61.5
		MCWB	53.5	53.0	58.1	62.9	68.6	73.6	76.3	75.2	72.2	67.6	60.2	56.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.2	65.1	67.0	71.4	75.5	78.6	81.5	80.9	78.0	75.4	70.1	68.4
		MCDB	70.0	72.5	74.9	80.0	85.2	89.2	92.6	91.2	84.1	80.5	74.2	73.1
	2%	WB	61.8	62.2	64.4	69.0	73.5	77.4	79.9	79.2	76.7	73.2	66.9	64.6
		MCDB	66.3	67.6	71.5	78.1	82.5	87.1	90.5	88.5	82.6	78.3	71.1	68.9
	5%	WB	58.0	58.3	62.0	66.9	72.0	76.3	78.4	78.0	75.6	71.2	64.0	61.4
		MCDB	62.7	64.2	69.3	75.3	80.1	85.2	88.2	86.1	80.9	76.3	68.2	65.1
	10%	WB	53.8	53.2	58.5	64.4	70.4	75.2	77.4	76.9	74.4	68.9	61.3	57.3
		MCDB	58.1	59.0	65.2	72.0	78.0	83.3	86.3	84.2	79.3	73.9	65.6	61.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.4	17.7	18.6	16.8	15.4	14.9	14.3	14.0	16.6	17.4	16.7
		MCDBR	22.1	24.7	24.9	23.6	20.8	18.7	17.8	17.2	17.1	18.5	20.4	20.3
		MCWBR	17.2	17.4	14.8	11.8	9.9	7.6	7.2	7.1	7.9	9.8	13.9	15.9
	5% WB	MCDBR	20.4	22.9	21.8	20.6	18.2	16.3	16.2	15.2	13.3	14.7	17.5	19.3
		MCWBR	17.5	18.6	16.2	12.6	9.7	7.5	7.6	7.0	6.9	8.9	14.2	16.8
(40) Clear-Sky Solar Irradiance	taub		0.312	0.323	0.349	0.392	0.437	0.483	0.511	0.504	0.409	0.358	0.327	0.316
	taud		2.542	2.505	2.463	2.353	2.261	2.181	2.118	2.125	2.384	2.517	2.551	2.559
	Ebn at Noon		281	291	291	282	270	256	248	247	268	274	273	272
	Edh at Noon		25	29	33	39	43	47	49	48	35	28	24	23
(44) All-Sky Solar Radiation	RadAvg		737	983	1320	1685	1899	2033	1938	1748	1430	1131	835	659
	RadStd		60	102	102	119	157	128	94	105	110	122	66	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.77	N/A	-2.16	N/A	N/A	N/A	N/A	+214	+164
(47) Regional (52 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	+299	+189

Nomenclature: See separate page